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CRYOGENIC RADIOMETRY: A NEW ABSORBER FOR X-RAYS UP TO 150 KeV

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Abstract

The accurate measurement of radiant power is essential for the calibration of X-ray detectors, such as silicon photodiodes. Cryogenic electrical substitution radiometers (ESRs) perform high accuracy and absolute measurements of radiant power. Material and geometry of the absorber in an ESR are chosen to maximize the absorption in the energy range of interest, while providing a high thermal response and a short time constant. The highest energy design previously reported allowed the measurement of X-rays up to 60 keV. In this work we present a new absorber developed at the Canadian Light Source for energies from 25 keV to 150 keV. Monte Carlo simulations led to a design with an absorption > 99% in the entire energy range considering all losses due to fluorescence and scattering. Measurements have been successfully performed at the Biomedical Imaging and Therapy (BMIT) beamline (05ID-2), which has a 3.7 T wiggler source and provides X-ray energies up to 140 keV.

INTRODUCTION

Cryogenic temperatures (typically around 4 K) provide the chance to measure low power beams by the physical principle of reducing the heat capacity of materials to very low values so a low power monochromatic incident beam absorbed by this material can provide a noticeable temperature variation that would be otherwise negligible at ambient temperatures. At this temperature, heat capacity and thermal noise of materials such as copper and gold dramatically reduce, enabling sensitivity and stability not possible to achieve at room temperature [1-3]. In this way, for achieving the highest-accuracy measurements of radiant power across the X-ray spectrum, the cryogenic absorber should be made to provide the highest possible absorption of the incident radiation power, so we can equate this incident X-ray radiant power (with precisely controlled electrical heating), directly to SI electrical units with minimal uncertainties. These features can be obtained by a combination of the appropriate material and geometrical design. In this way, ESRs function as absolute thermal detectors.

For high energies some material become transparent, e.g., copper at >30 keV, which makes them inefficient and inaccurate. In literature [1-3] can be found that, to overcome these limitations, researchers at PTB (Physikalisch-Technische Bundesanstalt) have designed a gold absorber with a thin copper shell for applications up to 60 keV. In this case Monte Carlo simulations were performed using Geant4 claiming a near-unity absorption with uncertainties well below 0.2%. Experimental measurements were performed at BESSY II using cavity absorbers of 500–550 µm gold base and ~80–90 µm copper shell. Authors claim standard uncertainties of less than 0.2–0.3% for 50 eV–60 keV photons

This work intends to push the limits forward by presenting a new absorber design able to be used up to 150 keV. For our design the same Monte Carlo approach is considered, but instead of Geant4, the FLUKA code is used. Simulation results show absorption levels > 99%.

DESIGN

The core concept involves the direct measurement of beam power, which can be used later to calculate the photon flux through the relationship:

$$P_{beam} = N * F * E. \quad (1)$$

Here P_{beam} is the beam power in J/s, N is the number of photons/s, F is the conversion factor $1.6*10^{-19}$ J/eV and E is the photon energy in eV.

Considering we are working with power levels in the range of microwatts (µW), reducing the heat capacity of the material is important so we can achieve noticeable temperature variations that can be accurately measured and controlled. As well, the highest absorption of the beam is required, as well as thermal isolation. For cryogenic conditions gold becomes a very good option. Its specific heat capacity drops from 0.129 J/g·K at room temperature to 0.00016 J/g·K at 4 K [4]. This is a reduction of 99.9%.

In a similar way, Gold also has good X-ray properties, particularly due to its high atomic number and known K- and L-edge behaviour which provides high absorption levels for a wide range of photon energies. In order to maximize X-ray absorption, the final geometric shape of the absorber was determined through iterative simulations and modelling. A conical geometry was adopted for the absorber, which provided the largest absorption of the beam considering all the possible interactions for the range analyzed, i.e., photoelectric absorption, Compton scattering, etc. As presented in Table 1, Monte Carlo simulations using the FLUKA code confirmed the absorber's efficiency across various energies. The design performs absorption levels > 99% for energies up to 150 keV, which means minimal loss with highest accuracy. Figure 1 shows some settings of the simulation.

The absorber is expected to remain at the same temperature when the beam is on and off. To achieve this condition an electrical resistor is attached to the absorber to compensate for the beam power when the beam is off, so the

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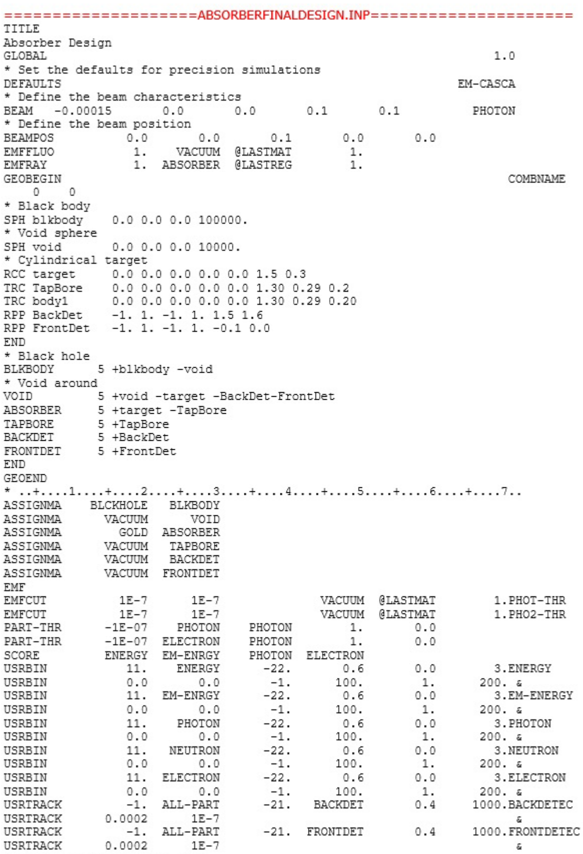
absorber's temperature remains constant during all the measurement process. This condition allows to accurately relate the electrical power P_{elec}=IV to the radiant beam power P_{beam}= P_{elec} Vacuum is required to maximize thermal isolation of the absorber to its surroundings. Finally, a PID (Proportional-Integral-Derivative) controller was implemented to maintain the absorber at a constant temperature to compensate the heat of the beam. A fine-tuning of the PID parameters was performed to ensure a rapid response to thermal changes, to keep thermal equilibrium and to minimize temperature fluctuations.

Table 1: Simulation results of the absorption level of the gold absorber for different beam energies.

Beam Energy (keV)	Absorption (%)
10	99.95
50	99.94
100	99.45
150	99.63



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DEFAULTS                                                          EM-CASCA
* Define the beam characteristics
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* Define the beam position
BEAMPOS          0.0       0.0       0.1       0.0       0.0
EMFFLUO           1.    VACUUM  @LASTMAT        1.
EMFRAY            1.  ABSORBER  @LASTREG        1.
GEOBEGIN                                                             COMBNAME
    0    0
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SPH blkbody    0.0 0.0 0.0 100000.
* Void sphere
SPH void       0.0 0.0 0.0 10000.
* Cylindrical target
RCC target     0.0 0.0 0.0 0.0 0.0 1.5 0.3
TRC TapBore    0.0 0.0 0.0 0.0 0.0 1.30 0.29 0.2
TRC body1      0.0 0.0 0.0 0.0 0.0 1.30 0.29 0.20
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RPP FrontDet   -1. 1. -1. 1. -0.1 0.0
END
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* Void around
VOID          5 +void -target -BackDet-FrontDet
ABSORBER      5 +target -TapBore
TAPBORE       5 +TapBore
BACKDET       5 +BackDet
FRONTDET      5 +FrontDet
END
GEOEND
* ..+....1....+....2....+....3....+....4....+....5....+....6....+....7..
ASSIGNMA    BLCKHOLE   BLKBODY
ASSIGNMA      VACUUM      VOID
ASSIGNMA        GOLD  ABSORBER
ASSIGNMA      VACUUM   TAPBORE
ASSIGNMA      VACUUM   BACKDET
ASSIGNMA      VACUUM  FRONTDET
EMF
EMFCUT          1E-7      1E-7                VACUUM  @LASTMAT        1.PHOT-THR
EMFCUT          1E-7      1E-7                VACUUM  @LASTMAT        1.PHO2-THR
PART-THR      -1E-07    PHOTON    PHOTON          1.       0.0
PART-THR      -1E-07  ELECTRON    PHOTON          1.       0.0
SCORE         ENERGY  EM-ENRGY    PHOTON  ELECTRON
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USRBIN           0.0       0.0       -1.      100.        1.      200. &
USRBIN           11.  EM-ENRGY      -22.       0.6       0.0        3.EM-ENERGY
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USRTRACK      0.0002      1E-7                                           &
USRTRACK         -1.  ALL-PART      -21.  FRONTDET       0.4     1000.FRONTDETEC
USRTRACK      0.0002      1E-7                                           &
* Set the random number seed
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Figure 1: FLUKA file showing the setup and some input cards of the simulations.

MEASUREMENTS

Measurements using this design have been performed at the BMIT-ID beamline of the Canadian Light Source (CLS). This beamline possesses a Wiggler and a Double Crystal Laue Monochromator of Silicon (1,1,1) crystals. Different filters and windows exist in the path of the beam from the source to the sample including air in the hutch. Some general information is presented in Table 2.

Table 2: Main featues of BMIT-ID beamline.

Feature	Description
Source type	Wiggler (26 periods, 48 mm length)
Field	up to 3.7 T
Critical energy	21 keV
Mono to source distance	44.5 m

Measurements have been taken up to 140 keV at two different fields of the Wiggler: 2.5 T and 3.7 T. Also, beam current was 220 mA for both cases. Details of the experimental set-up are shown in Fig. 2.

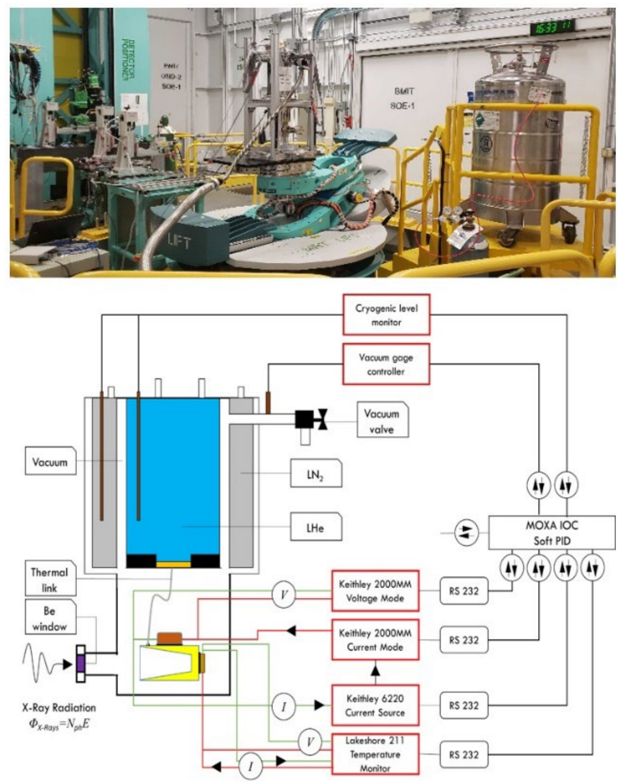

Figure 2: Experimental setup for monochromatic beam power measurement at the BMIT-ID (Canadian Light Source).

Measurements are plotted in Figs. 3 and 4 and compared to theoretical calculations that consider only transmission effects through all materials, i.e., diffraction at the monochromator is not considered in the theoretical calculations. For 2.5 T magnetic field strength a set of SiC (2.2 mm) and Cu (0.55 mm) filters was used, whereas for 3.7 T a set of SiC (2.2 mm) and Cu (1.1 mm) filters were used for the black squares, and an additional Cu (1.1 mm) was added for the red circles. In all cases the temperature sensor worked in the linear response region.

The results demonstrate an excellent agreement with the theoretical calculations for the 2.5 T case. This indicates that the assumption of 0.1% BW seems to be appropriate

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and beam diffraction at the monochromator is almost perfect. On the other hand, at 3.7 T larger discrepancies exist at higher energies. This means that the 0.1% BW assumption for the mono might not be appropriate. Also, the diffracted beam at the mono for higher fields and energies could reduce the flux. Other possible reasons could be beam misalignment due to positional shifts when changing energy or imperfect tuning of the monochromator's Bragg angles. In all cases errors were smaller than the size of the plotted data markers, indicating high precision and strong linearity across the energy spectrum. Besides, additional measurements were taken for 30, 60 and 100 keV for different beam power (filters configurations), showing a linearity of $R^2 > 0.98$ to theoretical filters transmission.

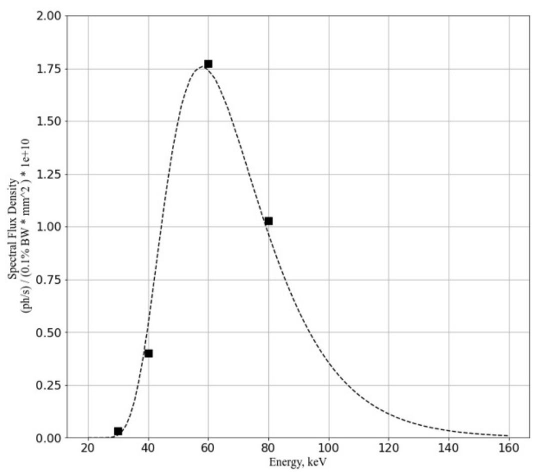

Figure 3: Measured photon flux using the radiometer (black squares) for wiggler at 2.5 T and theoretical calculations (dashed line).

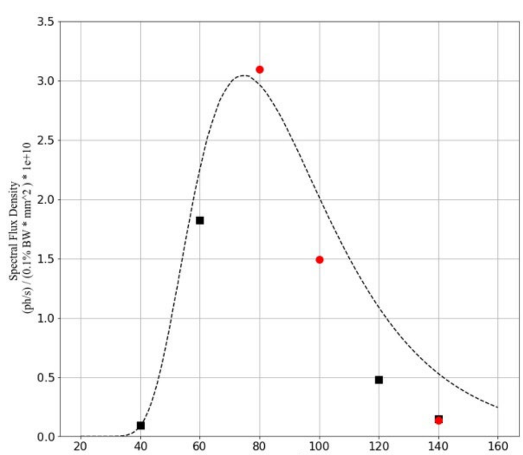

Figure 4: Measured photon flux using the radiometer (black squares and red circles) for wiggler at 3.7 T and theoretical calculations (dashed line).

It is worth mentioning that at 140 keV (3.7 T), absorber temperatures were tested at 17.5 K and 20 K obtaining absorbed powers of 79 μW and 83 μW respectively. At 30 keV (2.5 T), the absorbed power was measured in the order of 270 μW using different absorber temperatures and filter combinations. Besides BMIT [5], the radiometer has also been used for diodes calibration used for photon flux measurement at the Brockhouse (BXDS) beamline at the CLS [6].

CONCLUSION

The present work reports the design and implementation of a radiometer absorber. Monte Carlo simulations show absorption levels > 99% for energies up to 150 keV. The absorber is made of gold with a cylindrical shape and a conical tapered bore which is effective in minimizing transmission of the incident beam and maximize absorption of scattered beam. The radiometer showed a linear response due to the absorber cryogenic temperature and vacuum conditions, which allows to achieve an optimal thermal response. Measurements were taken up to 140 keV and showed good agreement with the theoretical transmitted flux calculation for 2.5 T of the Wiggler magnetic field. For 3.7 T and at higher energies (>=100 keV) discrepancies became relevant. As well, accuracy is below 1% thanks to the effectiveness of the PID-controlled temperature stabilization system.

These results highlight opportunities for further applications. The radiometer demonstrated to be a valuable tool for accurate measurement of photon flux, X-ray diagnostics and calibration efforts at the Canadian Light Source, and can be expanded to other synchrotron facilities.

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